

SMART INDIA HACKATHON

SIH 2026

ULTIMATE AI & HACKATHON TOOLKIT

"50+ tools to go from Problem Statement → Winning Prototype"

RESEARCH · IDEATION · DESIGN · ARCHITECTURE · DEVELOPMENT · TESTING ·
PRESENTATION

Use AI as your co-pilot, not as a replacement for your thinking.

How To Use This Toolkit

Smart India Hackathon rewards teams who move through the full journey deliberately — not teams who jump straight to coding. Here is the complete workflow this toolkit is organized around.

01 Understand the Problem

02 Research

03 Ideate

04 Validate the Solution

05 Design UI/UX

06 Create Architecture

07 Build Prototype

08 Test

09 Deploy

10 Prepare PPT

11 Practice Pitch

12 Final Demo

Don't use every tool. Pick the right tool for the job — the fastest team is not the one using the most tools, it's the one using the fewest tools well.

01

CATEGORY 01

Problem Statement & Research

Before you touch a single tool, understand the problem statement inside-out. These tools help you research faster — not think less.

TOOL	BEST FOR	SIH USE CASE	TIER
Perplexity	Quick, cited web research	Understand the problem domain fast with sources attached	FREE
ChatGPT	General reasoning & Q&A	Explain unfamiliar domains, summarize PS documents	FREE
Google Gemini	Research + Google integration	Cross-check facts, summarize long PDFs/reports	FREE
Claude	Deep reading & long documents	Digest lengthy government reports & PS annexures	FREE
Google Search	Ground-truth information	Find real-world context and existing news coverage	FREE
Google Scholar	Academic papers	Find peer-reviewed research related to the PS	FREE
NotebookLM	Source-grounded Q&A	Upload PS + reports, ask grounded questions only from them	FREE
Elicit	AI research assistant	Extract findings across many papers quickly	FREE
Consensus	Evidence-based answers	Check what research actually says on a claim	FREE
Semantic Scholar	Academic search engine	Find citations, related work, and authors	FREE
ResearchRabbit	Visual literature mapping	Discover related papers as a citation network	FREE

PRO TIP — Never blindly trust AI-generated research. Verify important facts, numbers and statistics using the original source before putting them on a slide.

02

CATEGORY 02

Ideation & Brainstorming

Your goal isn't to generate 100 ideas — it's to find the RIGHT one. Use these to widen your thinking, then narrow it down fast.

TOOL	BEST FOR	SIH USE CASE	TIER
ChatGPT	Rapid idea generation	Brainstorm feature lists, edge cases, alternate approaches	FREEMIUM
Claude	Structured reasoning	Build user personas, journeys, SWOT analysis	FREEMIUM
Gemini	Multimodal brainstorming	Brainstorm with images, docs and sheets together	FREEMIUM
Miro	Visual collaboration board	Team brainstorming, affinity mapping, journey maps	FREEMIUM
FigJam	Whiteboarding for teams	Live brainstorming sessions with the whole team	FREEMIUM
Notion AI	Docs + AI in one place	Organize ideas, meeting notes, and decisions centrally	PAID
Whimsical	Flowcharts & mind maps	Turn scattered ideas into structured mind maps	FREEMIUM

“Your goal isn’t to generate 100 ideas. Your goal is to find the RIGHT idea.”

03

CATEGORY 03

UI / UX Design

Design should communicate your solution instantly — even before you speak. Pick ONE primary tool and go deep rather than hopping between many.

TOOL	BEST FOR	SIH USE CASE	TIER
Figma	Professional UI design	Wireframes, high-fidelity UI, dev-ready screens	FREEMIUM
FigJam	Wireframing & flows	Quick low-fidelity user flows before detailed design	FREEMIUM
Canva	Fast visual design	Landing pages, social assets, quick mockups	FREEMIUM
Framer	Design-to-website	Interactive, publishable landing pages	FREEMIUM
Uizard	AI-assisted UI generation	Turn sketches/prompts into quick UI drafts	FREEMIUM
v0	AI-generated UI code	Generate React/Tailwind UI components from prompts	FREEMIUM
Lovable	Prompt-to-app builder	Turn a UI idea directly into a working app shell	FREEMIUM
Stitch	Google's AI UI design tool	Generate mobile/app UI concepts from text prompts	FREE

Wireframes • User flows • High-fidelity UI • Interactive prototypes • Landing pages • Dashboards • Mobile app screens

PRO TIP — Verify current pricing tiers before you publish or promise — free-tier limits on AI design tools change frequently.

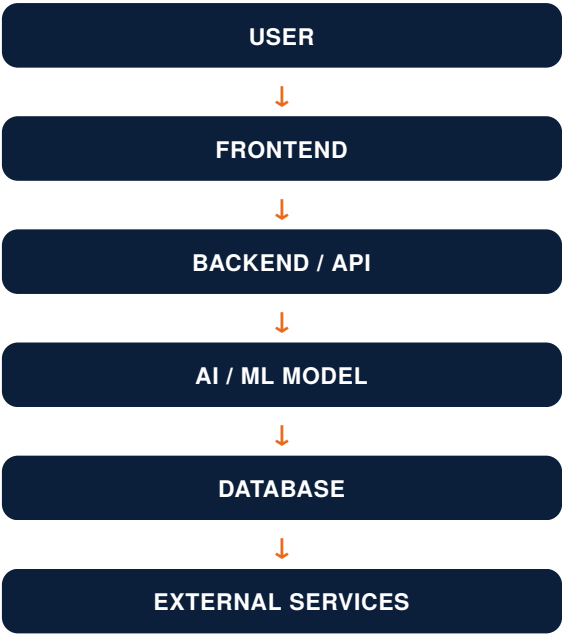
04

CATEGORY 04

System Architecture & Diagrams

Judges look for architecture that matches your ACTUAL implementation — not a generic diagram copied from the internet.

TOOL	BEST FOR	SIH USE CASE	TIER
Draw.io / diagrams.net	Free diagramming	System architecture, flowcharts, ER diagrams	FREE
Eraser	Dev-focused diagrams	Architecture diagrams generated from text/code prompts	FREEMIUM
Lucidchart	Professional diagramming	Detailed cloud & API architecture diagrams	FREEMIUM
Miro	Visual system mapping	Data flow diagrams, user journeys	FREEMIUM
Whimsical	Quick flowcharts	Fast sequence diagrams and flowcharts	FREEMIUM
Mermaid	Diagrams as code	Version-controlled flowcharts & sequence diagrams in README	FREE
Excalidraw	Hand-drawn style diagrams	Quick sketches for brainstorming architecture	FREE



“Your architecture diagram should match what you actually built — not what sounds impressive.”

05

CATEGORY 05

AI Coding & Development

AI can write code fast. It cannot understand your problem for you. Use it to accelerate — never to replace comprehension.

TOOL	BEST FOR	SIH USE CASE	TIER
Claude	Reasoning-heavy coding	Architect features, debug tricky logic, review code	FREEMIUM
ChatGPT	General-purpose coding help	Boilerplate, quick snippets, explaining errors	FREEMIUM
Cursor	AI-native code editor	In-editor AI pair programming across your whole codebase	FREEMIUM
GitHub Copilot	In-IDE autocomplete	Inline suggestions as you type in VS Code/JetBrains	FREEMIUM
Google Gemini	Coding + Google ecosystem	Code generation with strong Google Cloud integration	FREEMIUM
Antigravity	Agentic AI IDE (Google)	Autonomous agents that plan, code, test & verify tasks	FREE
Replit	Cloud IDE + AI agent	Build and run full projects in the browser instantly	FREEMIUM
Bolt.new	Prompt-to-full-stack-app	Scaffold a working full-stack app from one prompt	FREEMIUM
Lovable	Prompt-to-app builder	Rapidly build and iterate on a working app UI	FREEMIUM
v0	AI component generator	Generate production-ready UI components quickly	FREEMIUM

IMPORTANT — Never submit AI-generated code without understanding it. Judges may ask you to explain your implementation line by line.

06

CATEGORY 06

Backend, Database & APIs

Choose ONE backend stack your whole team understands — instead of stitching together five services nobody can debug on stage.

TOOL	BEST FOR	SIH USE CASE	TIER
Supabase	Postgres backend-as-a-service	Auth + database + storage + APIs in one place, fast setup	FREE MIUM
Firebase	Google's app backend platform	Realtime database, auth, hosting for mobile/web apps	FREE MIUM
MongoDB	NoSQL database	Flexible schema for rapidly changing prototype data	FREE MIUM
PostgreSQL	Relational database	Structured, relational data with strong consistency	FREE
Neon	Serverless Postgres	Instant, scalable Postgres with generous free tier	FREE MIUM
Appwrite	Open-source BaaS	Self-hostable backend with auth, DB, functions	FREE MIUM
Postman	API design & testing	Build, test and document your APIs collaboratively	FREE MIUM
Swagger / OpenAPI	API documentation standard	Auto-generate clean API docs for judges/dev team	FREE

“Choose ONE backend stack that your team understands instead of using too many services.”

07

CATEGORY 07

AI / ML Development

Before adding AI to your project, ask: does AI actually solve an important part of the problem — or is it decoration?

TOOL	BEST FOR	SIH USE CASE	TIER
Google AI Studio	Prototype with Gemini models	Quickly test prompts and multimodal AI ideas	FREE
Gemini API	Google's LLM API	Integrate Gemini models into your app	FREEMIUM
OpenAI API	GPT model API	Integrate GPT-based chat, vision, or reasoning features	PAID
Anthropic API	Claude model API	Integrate Claude for reasoning-heavy AI features	PAID
Hugging Face	Open-source ML hub	Find/host pretrained models, datasets, and spaces	FREEMIUM
Groq	Ultra-fast LLM inference	Low-latency AI responses for live demos	FREEMIUM
Replicate	Run ML models via API	Deploy vision/audio/image models without infra setup	PAID
Kaggle	ML notebooks & competitions	Experiment with models using free GPUs	FREE
Google Colab	Free cloud notebooks	Train/test ML models without local setup	FREEMIUM
Ollama	Run LLMs locally	Offline AI experimentation with no API cost	FREE

“Before adding AI, ask: Does AI actually solve an important part of the problem?”

08

CATEGORY 08

Datasets & Data Analysis

Good data beats a fancy model. Spend real time finding and cleaning the right dataset before you train anything.

TOOL	BEST FOR	SIH USE CASE	TIER
Kaggle	Datasets + notebooks	Find ready-to-use datasets across most domains	FREE
Google Dataset Search	Dataset search engine	Discover datasets indexed from across the web	FREE
Hugging Face Datasets	ML-ready datasets	Load datasets directly into training pipelines	FREE
Data.gov.in	Indian government open data	Official statistics for India-focused problem statements	FREE
Google Colab	Cloud notebooks	Explore, clean and visualize data collaboratively	FREEMIUM
Jupyter Notebook	Local notebook environment	Offline data exploration and model training	FREE

“For Indian / government-related SIH problems, always check official government datasets and reports first.”

09

CATEGORY 09

Testing & Debugging

A prototype that breaks on stage costs more than one missing feature. Test the boring paths, not just the happy path.

TOOL	BEST FOR	SIH USE CASE	TIER
Postman	API testing	Verify every API endpoint returns correct responses	FREEMIUM
Sentry	Error tracking	Catch and monitor runtime errors before judges do	FREEMIUM
Browser DevTools	Built-in browser debugging	Inspect network calls, console errors, layout issues	FREE
Jest	JavaScript testing framework	Unit test your core logic and functions	FREE
Playwright	End-to-end testing	Automate full user-flow testing across browsers	FREE
Cypress	End-to-end testing	Test critical user journeys in a real browser	FREEMIUM
Lighthouse	Performance & accessibility audit	Check speed, SEO and accessibility scores	FREE

MINI TESTING CHECKLIST

- ☐ API tested
- ☐ Authentication tested
- ☐ Invalid inputs tested
- ☐ Mobile UI tested
- ☐ Errors handled gracefully
- ☐ Loading states handled
- ☐ Full demo flow tested end-to-end

10

CATEGORY 10

Deployment & Cloud

Always try to have a working live demo — with a recorded backup video in case Wi-Fi fails on stage.

TOOL	BEST FOR	SIH USE CASE	TIER
Vercel	Frontend deployment	Deploy React/Next.js apps in seconds	FREEMIUM
Netlify	Frontend deployment	Simple static & JAMstack site hosting	FREEMIUM
Render	Full-stack app hosting	Deploy backend services and databases together	FREEMIUM
Railway	Full-stack app hosting	Quick backend + database deployment	FREEMIUM
Firebase	App hosting + backend	Deploy web/mobile apps with integrated backend	FREEMIUM
AWS	Enterprise cloud infrastructure	Scalable production-grade cloud deployment	PAID
Google Cloud	Enterprise cloud infrastructure	Cloud hosting with strong AI/ML integration	PAID
Microsoft Azure	Enterprise cloud infrastructure	Cloud hosting favoured by many SIH problem owners	PAID
Cloudflare	CDN, DNS & edge hosting	Speed, security and domain management	FREEMIUM

“Always have a working live demo if possible.”

11

CATEGORY 11

GitHub & Team Collaboration

A clean, well-documented GitHub repo signals a serious team — judges do check it.

TOOL	BEST FOR	SIH USE CASE	TIER
GitHub	Version control & hosting	Central code repository for the whole team	FREEMIUM
GitHub Projects	Task/board tracking	Track sprint tasks and progress visually	FREE
GitHub Issues	Bug & task tracking	Log bugs and feature requests during the hackathon	FREE
GitHub Actions	CI/CD automation	Auto-test and auto-deploy on every push	FREEMIUM
GitLab	Alternative Git platform	Version control with built-in CI/CD	FREEMIUM
Notion	Team documentation hub	Central place for docs, notes and task tracking	FREEMIUM

RECOMMENDED README.MD STRUCTURE

- └ Problem Statement
- └ Solution
- └ Features
- └ Architecture
- └ Tech Stack
- └ Installation
- └ Screenshots
- └ Demo
- └ Team
- └ Future Scope

12

CATEGORY 12

PPT & Presentation Design

Your PPT is often the FIRST thing judges evaluate. Clarity beats decoration every time.

TOOL	BEST FOR	SIH USE CASE	TIER
Canva	Fast, templated design	Polished slides, infographics and diagrams quickly	FREEMIUM
Gamma	AI-generated presentations	Turn an outline into a designed deck in minutes	FREEMIUM
PowerPoint	Industry-standard slides	Full manual control with Designer AI suggestions	PAID
Google Slides	Collaborative slides	Real-time team editing of the pitch deck	FREE
Beautiful.ai	Auto-designed slides	Slides that auto-adjust layout as you edit content	FREEMIUM

RECOMMENDED SIH PRESENTATION STRUCTURE

1. Problem
2. Current Situation
3. Proposed Solution
4. Innovation
5. Key Features
6. Technology Stack
7. Architecture
8. Workflow
9. Feasibility
10. Scalability
11. Impact
12. Prototype / Demo
13. Future Scope
14. Team

13

CATEGORY 13

Video, Demo & Content

Keep your demo video focused on Problem → Solution → Impact. Judges rewatch demo videos more than they reread PPTs.

TOOL	BEST FOR	SIH USE CASE	TIER
Canva	Quick video editing	Simple demo video edits and captions	FREEMIUM
CapCut	Mobile/desktop video editor	Fast-paced, polished demo video editing	FREE
Adobe Express	Quick creative content	Social clips, thumbnails, and short promos	FREEMIUM
Loom	Screen recording	Record product walkthroughs with narration	FREEMIUM
OBS Studio	Professional screen recording	High-quality local recording of your live demo	FREE
Descript	AI-powered video/audio editing	Edit video by editing the transcript text	FREEMIUM

“Keep your demo focused on the problem → solution → impact.”

14

CATEGORY 14

Pitch & Communication

The best solution with a weak pitch loses to an average solution with a clear one. Practise out loud, not just in your head.

TOOL	BEST FOR	SIH USE CASE	TIER
ChatGPT	Pitch refinement	Simplify technical explanations for non-technical judges	FREEMIUM
Claude	Structured pitch writing	Draft and tighten your 30-second elevator pitch	FREEMIUM
Gemini	Q&A practice partner	Generate likely judge questions to rehearse answers	FREEMIUM
Notion AI	Pitch documentation	Keep pitch scripts and Q&A prep organized as a team	PAID

JUDGES MAY ASK

- ▶ Why is this problem important?
- ▶ What makes your solution different?
- ▶ Why did you choose this technology?
- ▶ How does your AI model work?
- ▶ What happens if the system fails?
- ▶ Can this scale?
- ▶ What is the cost?
- ▶ Who will use it?
- ▶ How did you validate it?
- ▶ What is your future scope?

Category 15 — Final SIH Tool Stack

If you're overwhelmed by choice, here's a simple, proven default stack for a student team. Swap any row for a tool your team already knows well.

RESEARCH

Perplexity + Google Scholar + NotebookLM

IDEATION

ChatGPT + Miro

UI

Figma

ARCHITECTURE

Draw.io + Eraser

CODING

Cursor / Claude / GitHub Copilot

BACKEND

Supabase

AI

Gemini / OpenAI / Hugging Face

API TESTING

Postman

VERSION CONTROL

GitHub

DEPLOYMENT

Vercel / Render

PRESENTATION

Canva / Gamma

DEMO

Loom / OBS

This is a starting point, not a rule. The best stack is the one your team can confidently explain to a judge.

Don't Overuse AI

Tools improve your productivity. They do not replace problem understanding, innovation, implementation, validation, or presentation — the five things that actually decide the result.

AI CAN HELP YOU

- ✓ Research faster
- ✓ Brainstorm
- ✓ Code faster
- ✓ Debug
- ✓ Design
- ✓ Write documentation
- ✓ Prepare presentations
- ✓ Practice questions

BUT AI CANNOT REPLACE

- ✗ Understanding the problem
- ✗ Original thinking
- ✗ Teamwork
- ✗ Technical understanding
- ✗ Real-world validation
- ✗ Good implementation
- ✗ A convincing demo

USE AI TO MOVE FASTER. NOT TO THINK LESS.

SIH Final Checklist

Print this page. Go through it as a team the night before your final presentation.

PROBLEM

- ☐ Problem clearly understood
- ☐ Target users identified
- ☐ Existing solutions researched
- ☐ Problem is genuinely important

TECH

- ☐ Architecture ready
- ☐ Tech stack justified
- ☐ Prototype working
- ☐ APIs tested
- ☐ Database working
- ☐ AI integration working

FINAL

- ☐ Live demo tested
- ☐ Backup demo/video ready
- ☐ GitHub organized
- ☐ No broken links
- ☐ Everyone understands the project

SOLUTION

- ☐ Clear solution
- ☐ Unique value
- ☐ Realistic implementation
- ☐ Measurable impact

PRESENTATION

- ☐ PPT complete
- ☐ Architecture diagram ready
- ☐ Demo ready
- ☐ Team roles clear
- ☐ Judge questions prepared
- ☐ Pitch practiced

**BUILD SMART.
PRESENT CLEARLY.
SOLVE A REAL
PROBLEM.**

Problem → Research → Idea → Design → Build → Test → Deploy → Present → **IMPACT**

*“Tools don’t win hackathons.
Great problems + great ideas + strong execution do.”*